

MUMBAI SHIVAJI INTL, IndiaWMO: **430030**Lat: **19.089N**Lon: **72.868E**Elev: **11**StdP: **101.19**Time Zone: **5.50 (IND)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	18.8	5.8	5.7	26.1	7.9	6.6	26.7	6.3	26.9	5.9	27.5	0.6	50	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.0	36.0	22.7	35.0	23.0	34.1	23.4	28.0	31.2	27.6	30.8	27.2	30.5	3.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	30.0	26.9	22.5	29.8	26.2	21.7	29.3	90.2	31.5	88.4	31.1	86.8	30.8	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.3	6.5	6.1	15.1	39.2	2.3	1.4	13.4	40.2	12.1	41.0	10.8	41.7	9.1	42.7	
			DB												
			WB	10.9	29.0	1.6	0.7	9.8	29.4	8.8	29.8	7.9	30.2	6.7	30.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	25.2	26.1	28.2	29.4	30.4	29.4	28.0	27.9	28.1	29.5	28.6	26.5
	DBStd	2.18	2.09	2.35	1.94	1.41	0.99	1.68	1.20	1.04	1.13	1.51	1.59	1.91
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6613	471	452	565	581	632	582	558	556	543	603	558	512
	CDD18.3	3571	212	218	307	331	374	332	300	298	293	345	308	253
	CDH23.3	41537	1797	2101	3531	4199	5066	4257	3433	3343	3319	4337	3646	2508
	CDH26.7	17435	629	839	1587	1939	2600	1943	1097	996	1089	2041	1689	987
Wind	WSAvg	2.7	2.0	2.1	2.4	2.6	3.2	3.7	4.2	3.9	2.6	2.1	2.0	2.0
Precipitation	PrecAvg	2401	1	1	1	0	29	501	874	533	369	80	10	2
	PrecMax	3461	16	8	11	4	301	1020	1394	1005	897	371	83	24
	PrecMin	1479	0	0	0	0	0	74	219	172	85	0	1	0
	PrecStd	439	3	2	3	1	71	268	292	233	190	79	19	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	36.2	37.9	36.9	35.1	34.8	32.1	32.1	34.0	36.8	36.1	34.8
		MCWB	21.0	21.4	21.6	23.5	25.7	27.1	26.4	26.2	26.4	23.3	22.7	21.4
	2%	DB	32.3	34.2	35.8	35.0	33.9	33.4	31.2	31.1	32.1	35.2	34.9	33.2
		MCWB	20.4	21.1	21.6	23.9	26.4	26.5	26.4	26.0	25.9	23.3	22.3	21.2
	5%	DB	31.2	32.8	34.0	33.9	33.1	32.9	30.8	30.2	31.1	34.2	33.9	32.2
		MCWB	20.2	21.0	21.8	24.2	26.3	26.5	26.4	25.8	25.9	23.7	22.1	21.0
	10%	DB	30.1	31.1	32.8	32.8	32.8	32.1	30.0	29.9	30.2	33.1	33.0	31.1
		MCWB	20.0	20.6	22.1	24.5	26.1	26.4	26.2	25.8	25.6	24.3	22.3	20.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.3	24.5	27.1	27.9	28.4	28.5	27.8	27.5	27.9	28.2	26.7	25.3
		MCDB	29.1	30.2	31.0	32.4	32.6	31.6	29.6	29.5	30.6	31.6	31.1	29.9
	2%	WB	22.7	23.4	26.0	27.0	27.7	27.9	27.3	26.9	27.1	27.5	26.0	24.2
		MCDB	28.7	29.1	30.0	31.2	31.9	31.2	29.4	29.1	29.7	31.0	30.4	29.5
	5%	WB	21.9	22.7	25.2	26.4	27.3	27.6	27.0	26.6	26.7	26.9	25.4	23.3
		MCDB	28.2	28.7	29.7	30.7	31.5	30.9	29.2	28.8	29.2	30.6	30.0	28.8
	10%	WB	21.1	22.0	24.4	26.0	27.0	27.2	26.7	26.3	26.3	26.4	24.7	22.5
		MCDB	27.5	28.2	29.2	30.3	31.2	30.6	28.9	28.5	28.8	30.1	29.7	28.5
Mean Daily Temperature Range	5% DB	MDBR	10.7	10.3	9.1	7.0	5.0	4.5	3.8	3.7	4.8	7.7	9.8	10.6
		MCDBR	11.5	12.3	11.4	8.7	5.5	5.2	4.5	4.4	6.0	9.4	10.6	11.2
		MCWBR	5.1	5.4	5.3	4.2	2.6	2.6	2.5	2.2	2.6	4.3	4.6	4.9
	5% WB	MCDBR	10.6	10.0	8.2	6.5	4.8	4.7	3.9	3.7	4.9	6.8	8.2	9.6
		MCWBR	5.1	5.1	4.7	3.8	2.6	2.8	2.5	2.2	2.6	3.6	3.9	5.1
Clear-Sky Solar Irradiance	taub	0.453	0.452	0.507	0.548	0.557	0.664	0.689	0.629	0.570	0.588	0.532	0.492	
	taud	2.143	2.156	2.022	1.957	1.975	1.765	1.717	1.845	1.981	1.890	1.995	2.050	
	Ebn at Noon	819	846	812	781	765	680	665	710	749	715	737	764	
	Edh at Noon	144	150	177	191	185	226	238	210	182	193	166	154	
All-Sky Solar Radiation	RadAvg	4.97	5.83	6.69	7.18	7.08	4.74	3.52	3.77	4.66	5.24	4.93	4.71	
	RadStd	0.16	0.15	0.19	0.13	0.31	0.52	0.52	0.32	0.41	0.28	0.24	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.40	+1.68	-0.88	N/A	N/A	+0.33	N/A	N/A	+160	+161		
(47)	Regional (2 neighbors)	+0.34	+0.93	N/A	N/A	N/A	+0.22	N/A	N/A	+160	+161		

Nomenclature: See separate page